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EVEN SEMESTER EXAMINATION - 2024

Name of the Course: B. Tech (Mechanical Engineering) Semester: 4th

Name of the Paper: Manufacturing Execution System Paper Code: TMS-401

Time: 3 Hours Maximum Marks: 100

Note:-

(v) All questions are compulsory.

(vi) Answer any two sub questions among a, b & c in each main question.

(vii) Total marks in each main question are twenty.

(viii) Each question carries 10 marks.

Q1	(20marks)	**[CO1]**
(a)	Describe in detail about Production Systems. Also differentiate between Intermittent and Continuous Production systems.	
(b)	Highlight the differences between Mass Production system and Cellular Production system.	
(c)	Discuss in detail about the concept of 'Tomorrow's Manufacturing'. Also discuss in detail about Tomorrow's Manufacturing Plant.	
Q2	(20 marks)	**[CO1] [CO3] [CO5]**
(a)	Explain in detail about the concept of Enterprise Integration. Also state different types of integration involved in Enterprise Integration.	
(b)	Describe Capacity Requirement Planning (CRP)? Also state the strategies needed for its evaluation.	
(c)	Illustrate in detail about the concept of Manufacturing Process Management (MPM)? Also state its benefits.	
Q3	(20 marks)	**[CO3] [CO4]**
(a)	Explain in detail the concept of Internet of Things (IoT) alongwith its benefits.	
(b)	What do you understand by the term Information Sharing Infrastructure? Also state different benefits associated with it.	
(c)	Explain in detail about the concept of Industrial Internet of Things (IIoT) alongwith its benefits.	
Q4	(20 marks)	**[CO5]**
(a)	Describe in detail about the concept of Assembly Line alongwith its suitable benefits.	
(b)	Explain in detail about different types of wastes associated with today's Manufacturing plant.	
(c)	Discuss in detail about the concept of Line balancing? Also state different benefits associated with it.	
Q5	(20 marks)	**[CO6]**
(a)	Explain in detail about the term MES Standards alongwith their suitable applications.	
(b)	Explain in detail about the different steps involved in Assembly Line balancing with suitable examples.	
(c)	Describe in detail about the NAMUR MES Standard alongwith its benefits.	

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TME–403

B. TECH. (ME) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

MANUFACTURING PROCESS–II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the designation systems (ASA, ORS, and NRS) for the single point cutting tool with example. (CO1, CO2)

(b) Discuss tool wear and reasons responsible for tool wear in detail. (CO1, CO2)

(c) Discuss the tool life and Taylor's' tool life equation. If in turning of a steel rod by a given cutting tool at a given machining condition under a given machining environment, the cutting velocity increase from 30 m/mm to 80 m/mm and tool life decreases from 60 m 20 minute. Find the tool life at which cutting velocity will be 20 m/min. (CO1, CO2)

P. T. O.

2. (a) Discuss double housing planar machine with neat sketches. Also discuss reversible drive used in planar machines. (CO3)

   (b) What is drilling ? Discuss multi spindle drilling and gun drilling with neat sketches. (CO3)

   (c) Discuss function of a slotter machine with neat sketches. (CO3)

3. (a) Discuss planar type milling machine and differences between peripheral and face milling with neat sketches. (CO4)

   (b) Discuss the vertical spindle lapping machine with neat sketches. The designation of a grinding wheel is given as : 61-C-80-X-3-S-26. Explain the designation of the wheel. (CO4)

   (c) Do the compound indexing for 89 and 119 divisions. Also, do the differential indexing of 173 and 157 divisions. (CO4)

4. (a) Explain Abrasive Jet machining (AJM) and Electrodischarge machining (EDM) with neat sketches. (CO6)

   (b) What is part programming in CNC machines ? What are the meanings of N, G, S, F, and M codes ? Write 5G and 5M codes used in CNC machining. (CO6)

   (c) What is High Energy Rate Forging (HERF) ? Explain explosive forming with neat sketches. (CO6)

5. (a) Explain TIG and MIG welding. processes with neat sketches. (CO5)

   (b) Explain gas welding process and types of flames with neat sketches. (CO5)

   (c) Discuss chemical and electrochernical machining with neat sketches. (CO5)

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B. Tech
Name of the Paper: **Measurement and Instrumentation**
Time: 3 Hours

Semester: IV
Paper Code: TME 404
Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(20marks)**	CO 1/CO2
(a)	What are the principles of high precision measurements?	
(b)	Differentiate between accuracy and precision.	
(c)	Explain the principle and working of the vernier caliper and screw gauge. What is the importance of least count. Calculate the least count of any vernier caliper and screw gauge?	
Q2	**(20marks)**	CO 2/CO4/CO 5
(a)	What are the essential conditions for obtaining (a) Interference fits and (b) clearance fits?	
(b)	It is possible to drill a 25 mm nominal hole to an accuracy of 25± 0.02 mm using standard drill and drilling machine available. A shaft is to be machined to obtain a clearance fit in above hole such that allowance shall be 0.01 mm and maximum clearance should not be more than 0.08 mm. what should be the tolerance on the shaft.	
(c)	With the help of neat sketch explain the construction, working and applications of Optical Comparator. Also give the advantages and disadvantages of optical comparators.	
Q3	**(20 marks)**	CO 3 & CO 4
(a)	Explain the following methods for measuring effective diameter with the help of derivation (a)Two wire method (b) Three wire method	
(b)	Four end bars P, Q, R, S are to be calibrated using a calibrated length bar of 400 mm whose actual length is 399.9998 mm. The bar Q is longer than bar P by 0.0004mm, bar R is longer than bar P by 0.0003mm, while bar S is shorter than P by 0. 0001mm.The four gauges together have a combination length of 400.0002mm. Determine the corrected (actual) length of each end bar.	
(c)	Discuss the following terms Sensitivity, Hysteresis. Threshold, Pitch of gear, Calibration	
Q4	**(20 marks)**	CO3/CO4/CO5
(a)	What do you mean by subdivision of standards? Distinguish between primary, secondary, and working standards.	
(b)	Explain the terminology of screw thread with the help of neat sketch	
(c)	Explain hydraulic dynamometer ~~about~~ and discuss about different pressure gauges.	
Q5	**(20 marks)**	CO5/CO6
(a)	What do you mean by Analytical Balance? Derive the expression for its sensitivity.	
(b)	Explain the principle of interference. Explain how interference fringes are formed when optical flat is placed on a surface to be tested	
(c)	i. Write a short note on Cathode ray oscilloscope (CRO) ii. Differentiate between mechanical and electrical transducers	

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B.Tech (ME) Semester: IV

Name of the Paper: Kinematics of Machines Paper Code: TME 405

Time: 3 Hours Maximum Marks: 100

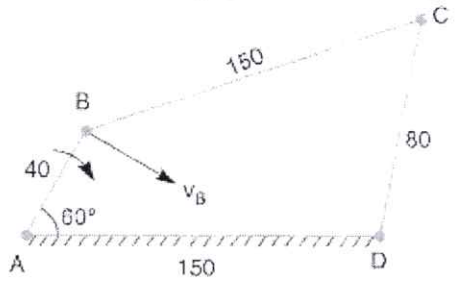
Note:

(i) All questions are compulsory

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each question are twenty

(iv) Each question carries 10 marks

Q1	(20 marks)	CO1
a)	Define the term "kinematics" and explain its significance in the theory of machines. Differentiate between "kinetics" and "kinematics" in the context of machine theory.	
b)	Describe the working principle of a four-bar linkage mechanism. Discuss its applications in real-world engineering systems.	
c)	Consider a mechanism consisting of four links connected by revolute joints. If each link has one degree of freedom, calculate the total number of degrees of freedom for the mechanism.	
Q2	(20 marks)	CO2
a)	Describe various inversions of a single slider crank mechanism with examples	
b)	For a four-bar mechanism with link lengths $a = 10\,\text{cm}$, $b = 15\,\text{cm}$, $c = 8\,\text{cm}$, and $d = 12\,\text{cm}$, determine the number of possible inversions.	
c)	Consider a Davis steering mechanism with a wheelbase of $2\,\text{m}$ and a track width of $1.5\,\text{m}$. If the inner and outer wheels are turned at angles of 20° and 15° respectively, calculate the turning radius of the inner and outer wheels.	
Q3	(20 marks)	CO3
a)	Discuss the principles of velocity analysis in mechanisms. Explain how velocity diagrams are used to analyse motion in linkages.	
b)	In a four bar chain ABCD as shown below, AD is fixed and is 150 mm long. The crank AB is 40 mm long and rotates at 120 rpm clockwise, while the link CD=80 mm oscillates about D. BC and AD are of equal length. Find the angular velocity of link CD when angle BAD=60°.  (All dimensions in mm).	

c)	What is the importance of acceleration analysis in the design of mechanical systems? Explain the process with suitable examples.	
Q4	(20 marks)	CO4
a)	State the Law of Gearing and explain its significance in gear systems. Define the following terms related to spur gears: i) Pitch circle ii) Module iii) Addendum iv) Dedendum v) Pressure angle	
b)	Explain the concept of a compound gear train. For a compound gear train consisting of two stages with gear ratios of 2:1 and 3:1 respectively, calculate the overall velocity ratio.	
c)	In an epicyclic gear train, if the sun gear has 20 teeth, the planet gears have 15 teeth each, and the internal gear has 60 teeth, calculate the velocity ratio.	
Q5	(20 marks)	CO5/CO6
a)	Describe the following types of cams: i) Plate or disk cam ii) Cylindrical cam Explain the different types of followers commonly used in cam mechanisms and their applications.	
b)	Given the desired follower motion as simple harmonic motion with an amplitude of 15 mm and a frequency of 3 Hz, draw a suitable cam profile.	
c)	Sketch the velocity and acceleration time curves for the following types of follower motions: a) Uniform velocity motion with a follower speed of 5 m/s. b) Simple harmonic motion with an amplitude of 10 mm and a frequency of 2 Hz. c) Cycloidal motion with a maximum velocity of 8 m/s and a dwell period.	

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End Semester Examination May/June 2024

Name of the Course: **B.Tech** Semester: **IV**

Name of the Paper: **Mechanics of Materials** Paper Code: **TME 406**

Time: 3 Hours **Maximum Marks: 100**

Note:

(i) All questions are compulsory

(ii) Answer any two sub questions among a, b &c in each main question

(iii) Total marks in each question are twenty

(iv) Each question carries 10 marks

Q1	**(20 marks)**	**CO1, CO2**
(a)	What is Mohr's stress circle? How the Mohr's circle is constructed?	
(b)	Deduce expression to determine the elongation of a bar of tapering circular section.	
(c)	What do you mean by thermal stresses? Explain	
Q2	**(20 marks)**	**CO3**
(a)	What do you mean by shear force and bending moment diagram? Explain the procedure of drawing SFD and BMD with example.	
(b)	A simply supported beam has distributed a load of linearly varying intensity with zero at one end to w per unit run at the other. Draw the shear force and bending moment diagrams. Also find maximum bending moment produced.	
(c)	A cantilever beam of 10m span carries loads of 4 kN and 6 kN at 2m and 6m respectively from the fixed end along with another load of 6 kN at free end. Draw the shear force and bending moment diagram.	
Q3	**(20 marks)**	**CO4**
(a)	Deduce the torsion equation stating the assumptions made.	
(b)	A hollow steel shaft transmits 200 kW of power at 150 rpm. The total angle of twist in a length of 5 m of the shaft is 3°. Find the inner and outer diameters of the shaft if the permissible shear stress is 60 MPa. G = 80 GPa.	
(c)	A 120 mm wide and 10 mm thick steel plate is bent into a circular arc of 8 m radius. Determine the maximum value of stress produced. Also find the bending moment which will produce the maximum stress? E = 200 GPa.	
Q4	**(20 marks)**	**CO4, CO5**
(a)	Derive the expression for maximum slope and maximum deflection of a simply supported beam subjected to uniformly distributed load over the whole span of the beam.	
(b)	A cast-iron pipe of 750-mm diameter is used to carry water under a head of 60 m. Determine the thickness of the pipe if the permissible stress is to be 20 MPa.	
(c)	A simply supported beam has a span of 15 m and carries two point loads of 4 kN and 9 kN at 6 m and 10 m respectively from one end. Find the deflection under each load and maximum deflection. Take E= 200 GPa and MOI(I) = 400×10^6 mm^4.	

Q5	(20 marks)	
(a)	What is meant by crippling or buckling load? Derive the formula for Euler's crippling load for a column subjected to one end fixed and other end free, end conditions.	CO5, CO6
(b)	What assumptions are taken in the analysis of thin cylinders? Deduce the expression for circumferential and longitudinal stresses developed in them.	
(c)	A 3.2 m long fixed-end hollow cast-iron column has its internal and external diameters as 60 mm and 80 respectively. Determine Rankine's crippling load using the value of crushing stress to be 500 MPa and the value of the Rankine's constant 1/1600.	

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B.Tech | Semester: IV

Name of the Paper: Fluid Mechanics and Fluid Machines | Paper Code: **TME 407**

Time: 3 Hours | Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	(20marks)	**CO1**
(a)	Apply Bernoulli equation to a venturi meter with diagram.	
(b)	Apply Bernoulli equation to an orifice meter with diagram.	
(c)	Explain dimensional homogeneity?	
Q2	(20 marks)	**CO1,CO2**
(a)	What is Froude number and where it can be applied?	
(b)	Show that the shear stress in a steady laminar flow in a circular pipe is linear.	
(c)	What are major and minor losses? How can major loss be estimated?	
Q3	(20 marks)	**CO2,CO3**
(a)	What are Pi terms in Buckingham Pi theorem? How do you choose the variables for the Pi terms?	
(b)	Explain how dimensional analysis can be beneficial for comparing the performance of scale model with the prototype.	
(c)	What is Reynolds number? How does the flow characteristics change with Reynolds number, explain with diagram.	
Q4	(20 marks)	**CO3,CO4**
(a)	What is a boundary layer? What is the significance of its displacement thickness?	
(b)	Explain stress tensor in a fluid. How is it related to the velocity field?	
(c)	How will the head loss in a pipe change if its diameter is reduced to half?	
Q5	(20 marks)	**CO5,CO6**
(a)	What will be the force exerted by a jet striking a moving curved vane?	
(b)	Under what condition can a Pelton turbine achieve maximum efficiency?	
(c)	Explain the various parts of a centrifugal pump.	

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: **B.Tech** Semester: **IV**
Name of the Paper: **Applied Thermodynamics** Paper Code: **TME408**

Time: **3 Hours** Maximum Marks: **100**

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Draw the air standard Otto cycle on p-v and T-s diagram and derive an expression of its thermal efficiency.	**CO2, CO4**
(b)	In a test of a single-cylinder, four stroke Diesel engine, the following data were recorded: Indicated mean effective pressure = 755 kPa cylinder diameter = 10 cm piston stroke = 15 cm engine speed = 480 rpm brake wheel diameter = 62.5 cm net load on the brake wheel = 170 N Calculate: (a) indicated power, (b) brake power, and (c) the mechanical efficiency of the engine.	**CO3, CO6**
(c)	An air-standard Diesel cycle has a compression ratio of 14. The pressure at the beginning of the compression stroke is 1 bar and the temperature is 300 K. The maximum cycle temperature is 2500 K. Determine the cut-off ratio and thermal efficiency.	**CO3, CO6**
Q2	(20 marks)	
(a)	Derive an expression for the thermal efficiency of an air standard Brayton cycle. Also explain how pressure ratio of the cycle affects the efficiency and net workdone.	**CO2, CO4**
(b)	Explain the principle of a propulsion systems, also classify them. Explain the working of any three propulsive engine with the help of diagrams.	**CO1, CO2**
(c)	The pressure ratio and maximum temperature of an air standard Brayton cycle are 6:1 and 950 K, respectively. Air enters the compressor at 1 bar and 300 K. Calculate for 1 kg of air flow, the compressor work, turbine work and efficiency of cycle.	**CO3, CO6**
Q3	(20 marks)	
(a)	What are the four basic components of a steam power plant, write their functions in brief? Also explain why Carnot vapour power cycle is not practical for a steam power plant.	**CO1, CO2**
(b)	Explain the method of reheating and regeneration in Rankine vapour power cycle. Also discuss the effects of these methods on the cycle performance.	**CO3, CO6**

(c) A steam power plant operates on an ideal Rankine cycle. The steam enters the turbine at 3 MPa, 300°C and is condensed in the condenser at a pressure of 50 kPa. Use below given tables and determine the heat supplied, thermal efficiency, back work ratio and work ratio of this cycle. At 3 MPa, the saturation temperature of the water is 233.85°C. **CO3, CO6**

Saturated water—Pressure table

		Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
Press., P kPa	Sat. temp., T_{sat} °C	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
50	81.32	0.001030	3.2403	340.49	2142.7	2483.2	340.54	2304.7	2645.2	1.0912	6.5019	7.5931

Superheated steam table:

3 MPa				
T (°C)	v (m^3/kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg.K)
300°C	0.08118	2750.8	2994.3	6.5412

Q4 (20 marks)

(a) Explain with a diagram how does a reciprocating compressor increase the pressure of air. Also, derive an expression for its work done by neglecting clearance. **CO3, CO5**

(b) (I) Define the volumetric efficiency of a compressor. How is it affected by (i) pressure ratio, & (ii) clearance ratio? **CO1, CO5**

(II) Discuss the significance of intercooling on the performance of multi-stage compression.

(c) A single stage, single acting reciprocating air compressor has a swept volume of 2000 cm^3 and runs at 800 rpm. It operates on a pressure ratio of 8. The compressor has a clearance of 5% of the swept volume. The suction conditions are at 0.97 bar and 27°C. The room conditions are at 1.01 bar and 17°C. The polytropic compression and expansion with n = 1.25. Calculate: **CO3, CO5**

(i) Indicated power,
(ii) Volumetric efficiency,
(iii) FAD.

Q5 (20 marks)

(a) Explain the working of a vapour absorption refrigeration system with the help of a neat diagram. **CO2, CO4**

(b) Define the following terms: **CO1**

(i) Refrigeration,
(ii) Air conditioning,
(iii) COP,
(iv) Refrigeration effect (RE),
(v) Ton of Refrigeration (TR).

(c) Explain the following psychrometric terms/processes with the help of a psychrometric chart. **CO1, CO2, CO4**

(i) Wet bulb temperature
(ii) Dew point temperature
(iii) Sensible heating and cooling
(iv) Humidification and dehumidification
(v) Enthalpy of moist air